📘 Database Notes

ACID Properties

ACID is an acronym for four key properties that ensure reliable database transactions.

1. A – Atomicity

> A transaction is **all or nothing**.
> If any part of the transaction fails, the entire transaction is rolled back.

Example: If you're transferring money from Account A to B, both debit and credit operations must succeed — or neither.

2. C – Consistency

> The database must remain in a **valid state** before and after a transaction.
> It ensures that a transaction doesn't violate any **constraints, rules, or relationships**.

3. I – Isolation

> Multiple transactions can occur at the same time **without affecting each other**.
> Intermediate steps of a transaction are **not visible** to other operations.

4. D – Durability

> Once a transaction is committed, the changes are **permanent**, even in the case of a system failure.

Example: After placing an order online, even if the system crashes, your order still goes through.

Data Redundancy

Definition:
Data redundancy occurs when **the same piece of data is stored in multiple places** within a database.

Example:
If a student's name and address are stored in both the **Students** table and the **Course Enrollment** table, it leads to redundancy.

Problems caused by redundancy:

- Wastes storage space
- Makes updates difficult
- Increases the chance of **data inconsistency**

How to reduce it:

- Apply **Normalization** (1NF, 2NF, 3NF, etc.)
- Use foreign keys and separate related data into different tables

---

Data Integrity

Definition:
Data integrity ensures that the **data is accurate, consistent, and reliable** throughout its lifecycle in a database.

Types of Data Integrity:

1. **Entity Integrity** – Each table must have a unique primary key.
2. **Referential Integrity** – Foreign keys must match primary keys in the referenced table.
3. **Domain Integrity** – Data values must fall within defined rules (like data type, format, etc.).
4. **User-defined Integrity** – Custom rules defined by the user.

Why it's important:

- Prevents errors and corruption
- Ensures that relationships between tables remain valid

- Maintains **trustworthy data**